Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2024-P1-Model)** **Date: 04-01-2026**

Time: 09.00Am to 12.00Pm **GTA-10** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**D**	2	**D**	3	**D**	4	**B**	5	**BD**	6	**ABCD**
7	**ABD**	8	**4**	9	**6**	10	**1**	11	**8**	12	**3**
13	**3**	14	**A**	15	**D**	16	**A**	17	**C**		

PHYSICS

18	**D**	19	**C**	20	**A**	21	**C**	22	**ABD**	23	**AB**
24	**ACD**	25	**8**	26	**2**	27	**5**	28	**5**	29	**3**
30	**3**	31	**C**	32	**B**	33	**A**	34	**A**		

CHEMISTRY

35	**C**	36	**A**	37	**A**	38	**C**	39	**ABD**	40	**BCD**
41	**ABD**	42	**3**	43	**31**	44	**2**	45	**4**	46	**8**
47	**3**	48	**C**	49	**D**	50	**A**	51	**A**		

For More Join: @IITJEE_Advanced

SOLUTIONS
MATHEMATICS

1. In triangle ABC $\cos^2 A+\cos^2 B+\cos^2 C+2\cos A\cos B\cos C=1$

$\cos^2 A+\cos^2 B+2\sin A\sin B\cos C=\frac{15}{8}$ _____(1)

$\cos^2 B+\cos^2 C+2\sin B\sin C\cos A=\frac{14}{9}$ _____(2)

Add $\cos^2 C+2(\cos A\cos B-\sin A\sin B)\cos C$ to both sides of the first given equation.

$1=\frac{15}{8}-2\cos^2 C+\cos^2 C$ so cosC is $\sqrt{\frac{7}{8}}$ and therefore sin C is $\sqrt{\frac{1}{8}}$.

Similarly, we have cos A = $\sqrt{\frac{14}{9}-1}=\sqrt{\frac{5}{9}}$

2. $P\left(4,\frac{-26}{3},\frac{-10}{3}\right)$

$\overline{AB}=\hat{i}+\hat{j}+3\hat{k},\overline{AC}=-\hat{i}-\hat{j}+2\hat{k},\overline{BC}=-2\hat{i}-\hat{k}$ $\overline{AC}\perp\overline{BC}$

$\overline{AC}\perp\overline{BC}$

B D

C A

$\frac{A+B}{2}=\frac{C+D}{2}\Rightarrow D=(4,2,0)$

Equation of plane through A, B, C, D is $x-5y-2z+6=0$ foot of perpendicular drawn form P is (h, k, r).

$\frac{h-4}{1}=\frac{k+\frac{26}{3}}{-5}=\frac{r+\frac{10}{3}}{-2}=\frac{-\left(4+\frac{130}{3}+\frac{20}{3}+6\right)}{30}\Rightarrow \quad (h,k,r)=\left(2,\frac{4}{3},\frac{2}{3}\right)=(\ell,m,n)$

$\ell+m-2n=2,\quad \ell>m>n$ volume of pyramid $=\left(\frac{1}{3}\right)$(Base Area) (height)

$=\left(\frac{1}{3}\right)\left(\sqrt{30}\right)\left(\sqrt{120}\right)=20$ cubic units

3. S_2 gets R_2 and remaining 5 students does not get the seat allotted to him/her is $D_5=44$.

$n(T_1\cap T_2\cap T_3\cap T_4\cap T_5)$ = Total $-\ n\left(\overline{T_1}\cup\overline{T_2}\cup\overline{T_3}\cup\overline{T_4}\cup\overline{T_5}\right)$

$=6!-n\left(\overline{T_1}\cup\overline{T_2}\cup\overline{T_3}\cup\overline{T_4}\cup\overline{T_5}\right)$

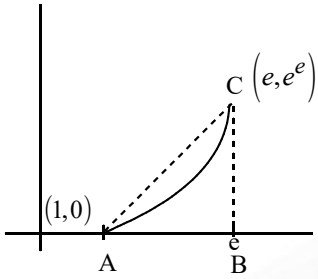
$=6!-[5(5!)(2!)-4(4!)(2)-6(4!)(2!)(2)+3(3!)(2)+5(3!)(2!)(2!)(2)+3!(2!)(2!)(2!)-2(2!)2-3(2!)(2!)2+2]$

4. $f(y)=(\ln x)e^x$

$f'(x)=e^x\left(\ln x+\frac{1}{x}\right)\quad f''(x)=e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)+e^x\left(\ln x+\frac{1}{x}\right)-e^x\left(\ln x+\frac{2}{x}-\frac{1}{x^2}\right)$

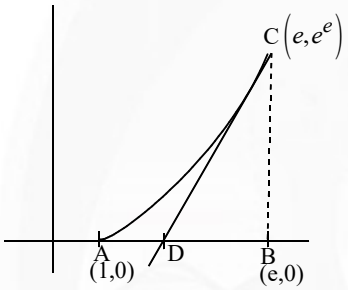
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$x \in (1,e) \qquad f''(x) > 0, f'(x) > 0$



$$\int_1^e (\ln x)e^x dx < \text{Area of } \Delta \ell e \text{ABC} < \left(\frac{1}{2}\right)(e-1)\left(e^e\right) < \frac{e^{e+1}-e^e}{2}$$

I > Area of the triangle formed by tangent drawn at (e,e^e)



equation of tangent at C is $y - e^e = e^e\left(1+\frac{1}{e}\right)(x-e) \qquad D\left(\frac{e^2}{e+1},0\right)$

$$\text{Area of } \Delta \ell e \text{BCD} = \frac{1}{2} \times \left(e - \frac{e^2}{e+1}\right)e^e = \frac{e^{e+1}}{2(e+1)} \qquad I > \frac{e^{e+1}}{2(e+1)}$$

5. $$\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4} \qquad \text{_____ (1)}$$

$$\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right) = \frac{\pi}{4} + \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)$$

$$\frac{x+1}{\sqrt{(x+1)^2+1}} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{x^2+1}} + \frac{1}{\sqrt{2}} \cdot \frac{x}{\sqrt{x^2+1}} = \frac{x+1}{\sqrt{2}\left(\sqrt{x^2+1}\right)}$$

$$(x+1) = 0 \text{ (or) } \sqrt{x^2+2x+2} = \sqrt{2x^2+2}$$

x = –1 (or) $x^2 = 2x \Rightarrow x = 0, 2$

x = 2 will not satisfy _____ (1)

S = {0, –1}

$$\sum_{x \in S} \sin\left(\frac{\pi}{2}\left(x^2+x+5\right)\right) - \cos\left(\pi\left(x^2+x+5\right)\right) = \left(\sin\frac{5\pi}{2} - \cos 5\pi\right) + \left(\sin\frac{5\pi}{2} - \cos 5\pi\right)$$

= 2 + 2 =4

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6. $\log_a b = t \Rightarrow t + \frac{6}{t} = 5 \Rightarrow t^2 - 5t + 6 = 0 \Rightarrow t = 2,3$

i) $\log_a b = 2 \Rightarrow b = a^2$ $45^2 = 2025$ and $b \le 2026 \Rightarrow a_{\max} = 45$

ii) $\log_a b = 3 \Rightarrow b = a^3$ $13^3 = 2197$ and $b \le 2026 \Rightarrow a_{\max} = 12$

number of ordered pairs possible are $44 + 11 = 55$

$\boxed{R = 55}$

7. Consider $A = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix} = I + B$

$$B^2 = \begin{bmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & ac \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & ac \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^{2025} = (I+B)^{2025} = I + {}^{2025}C_1 B + {}^{2025}C_2 B^2 + {}^{2025}C_3 B^3 + ----$$

$$= I + 2025B + {}^{2025}C_2 B^2 = \begin{bmatrix} 1 & 2025a & 2025b + 2025C_2 ac \\ 0 & 1 & 2025c \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{2025} + A = 2I + 2026B + {}^{2025}C_2 B^2$$

$$\begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2026a & 2026b + {}^{2025}C_2 ac \\ 0 & 2 & 2026c \\ 0 & 0 & 2 \end{bmatrix}$$

$$2026a = 2 \Rightarrow a = \frac{1}{1013} \qquad 2026c = 2 \Rightarrow c = \frac{1}{1013}$$

$$2026b + {}^{2025}C_2 ac = 0 \Rightarrow b = \frac{-{}^{2025}C_2 ac}{2026} = \frac{-(2025)(2024)}{(2)(2026)}\left(\frac{1}{1013}\right)^2 = \frac{-(2025)(1012)}{2(1013)^3}$$

$$Tr\left(A^{2025}\right) = 3 \qquad \left|A^{2026}\right| = 1$$

8. $$\int_{2a}^{8a} \frac{1}{\left(f^{-1}(x)\right)^2 + \left(f^{-1}(x)\right)^4}$$

Consider $f^{-1}(x) = t \Rightarrow x = f(t) \Rightarrow dx = f'(t)dt$

$f^{-1}(2a) = \alpha, \qquad f^{-1}(8a) = \beta$

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$$I=\int_{\alpha}^{\beta}\frac{4t^3+t^2+2t}{t^2+t^4}dt=\int_{\alpha}^{\beta}\frac{4t^3+2t}{t^4+t^2}dt+\int_{\alpha}^{\beta}\frac{t^2}{t^2+4}dt=\log_e\left(t^4+t^2\right)\Big|_{\alpha}^{\beta}+\int_{\alpha}^{\beta}\frac{dt}{t^2+1}$$

$$I=\log_e\left(\frac{\beta^4+\beta^2}{\alpha^4+\alpha^2}\right)+\tan^{-1}(\beta)-\tan^{-1}(\alpha)$$

$$f^{-1}(2a)=\alpha \qquad \Rightarrow 2a=f(\alpha)=\alpha^4+\frac{\alpha^3}{3}+\alpha^2$$

$$f^{-1}(8a)=\beta \qquad \Rightarrow 8a=f(\beta)=\beta^4+\frac{\beta^3}{3}+\beta^2$$

$$I=\log_e\left(\frac{8a-\frac{\beta^3}{3}}{2a-\frac{\alpha^3}{3}}\right)+\tan^{-1}(\beta)-\tan^{-1}(\alpha)$$

$a\to\infty \qquad \Rightarrow \alpha,\beta\to\infty \qquad I=\log_e 4$

9. Since MN is tangent to C_1 at M, $\angle NMQ=\angle MPQ$. Since MN = PN, ΔMNP is isosceles so $\angle MPN=\angle PMN$. It follows that $\angle NPQ=\angle PMQ$. But MN is tangent to C_2 at N, so $\angle NPQ=\angle MNQ$. Hence, $\angle MNQ=\angle PMQ$. Combining this with the fact that $\angle NMQ=\angle MPQ$, we see that $\Delta PMQ\sim\Delta MNQ$. Then $\frac{PQ}{QM}=\frac{QM}{QN}$, so

$QM^2=PQ.QN=3.2=6$

10. $Z=a+b\omega+c\omega^2$

$Z^{2023}=\overline{Z}\Rightarrow|Z|=$ o (or) 1

$|Z|=0\Rightarrow a+b\omega+c\omega^2=0 \qquad \Rightarrow a=b=c$

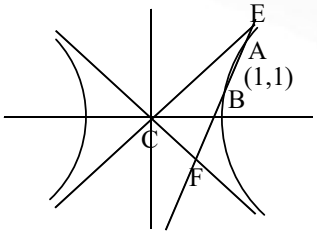
$|Z|=1 \qquad \Rightarrow \qquad Z\overline{Z}=1$

$\Rightarrow \qquad \left(a+b\omega+a\omega^2\right)\left(a+b\omega^2+c\omega\right)=1$

$\Rightarrow \qquad a^2+b^2+c^2-ab-bc-ca=1$

11. Equation of chord AB is $S_1=S_{11}$ whose mid point is (1, 1)

$\Rightarrow 7x(1)+4\left(x(1)+(y)(1)\right)-y(1)-4=7+8-1-4 \Rightarrow 11x+3y=14$



Equation of circumcircle CEF is $x^2+y^2-ax-by+c=0$

Homogenising circumcircle with the line EF the asymptotes are obtained.

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$$x^2+y^2-ax(1)-by(1)+c(1)^2=0$$

$$x^2+y^2-ax\left(\frac{11x+3y}{14}\right)-by\left(\frac{11x+3y}{14}\right)+c\left(\frac{11x+3y}{14}\right)^2=0$$

$$\Rightarrow x^2\left[1-\frac{11a}{14}+\frac{121c}{196}\right]+y^2\left[1-\frac{3b}{14}+\frac{9c}{196}\right]+xy\left[\frac{-3a}{14}-\frac{11b}{14}+\frac{66c}{196}\right]=0$$

Equation of asymptotes is $7x^2+8xy-y^2=0$

$$\frac{1-\frac{11a}{14}+\frac{121c}{196}}{7}=\frac{1-\frac{3b}{14}+\frac{9c}{196}}{-1}=\frac{\frac{-3a}{14}-\frac{11b}{14}+\frac{66c}{196}}{8}$$

$a=0,\ a=\frac{112}{23},\ b=\frac{64}{23}$ $\quad \frac{1-\frac{11a}{14}}{7}=\frac{1-\frac{3b}{14}}{-1}$ $\quad \Rightarrow 11a+21b=112$

12. $g^1(x)=x^2\int_1^x f(u)du$

$$g^1(1)=0, g^{11}(x)=x^2f(x)+\int_1^x f(u)du.2x \Rightarrow g^{11}(1)+g^1(1)=f(1)+0=3$$

13. $z=m+i\sqrt{n},\ \bar{z}=m-i\sqrt{n}$

$z,\bar{z}$ are roots of x^3+ax+b and x^3+cx^2+d

$z+\bar{z}-20=0 \quad 2m-20=0 \quad m=10$

$-21,\ z,\bar{z}$ are roots of x^3+cx^2+d

$-21z-21\bar{z}+z\bar{z}=0 \quad -21\left(z+\bar{z}\right)+|z|^2=0$

$\Rightarrow |z|^2=(21)(20) \Rightarrow m^2+n=420 \Rightarrow n=320$ $\quad \frac{n-2m}{100}=\frac{320-20}{100}=3$

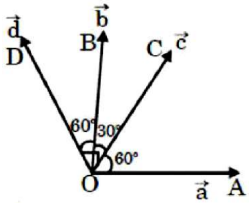
14. P) Mean $\frac{100+5k}{18+k}\Rightarrow$ if $k=7$ mean $=5.4$

Q) $\frac{100+5k}{18+k}=5.5\Rightarrow k=2$

R) $\bar{x}=\frac{110}{20}=5.5 \quad \therefore$ Mean deviation about

$$\text{mean}=\frac{(4\times3.5)+(8\times1.5)-(3\times3.5)+(2\times0.5)+(3\times2.5)}{20}=2.25$$

S) $\frac{100+5k}{18+k}=5.25\Rightarrow k=22 \therefore \text{Mode}=5$

15. 

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($\hat{p}$ is unit vector $\perp$ to the plane) $\sin\theta_1\sin\theta_2\cos\phi=1,\ \cos\alpha=\frac{1}{2}$

$\theta_1=\theta_2=90^\circ,\phi=0,\alpha=\frac{\pi}{3}\ \Rightarrow$ OABCD are coplanar

$\Delta ABC=\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right| =\frac{1}{2}\left|2\hat{p}-3\hat{p}-\frac{3\sqrt{3}}{2}\hat{p}\right|=\frac{1}{2}\left(1+\frac{3\sqrt{3}}{2}\right)$

16. Clearly $I_1+I_2=\sqrt{2}\,e^{\pi/4}-1$ and $I_3=\frac{1}{2}$

17. A) $\left(\alpha^3+1\right)\left(\beta^3+1\right)\left(\gamma^3+1\right)\left(\delta^3+1\right)$

$T1(\alpha+1)(\alpha+\omega)\left(\alpha+\omega^2\right)=f(-1)f(-\omega)f\left(-\omega^2\right)$

$=(4)\left(2\omega^2+2\right)(2\omega+2)=(4)(4)=16$

B) $\prod_{k=1}^{179}\sin k^\circ=\sin1^\circ\sin2^\circ-\text{-------}\sin179^\circ\, z^{180}=1$

$z^{180}-1=(z-1)(z-\alpha)-\text{-----}\left(z-\alpha^{179}\right)$

$\frac{z^{180}-1}{z-1}=(z-\alpha)\left(z-\alpha^2\right)-\text{------}\left(z-\alpha^{179}\right)$

$\Rightarrow\quad (180)=(12)^{179}\sin1^\circ\sin2^\circ-\text{-----}\sin179^\circ$

$\sin1^\circ\sin2^\circ-\text{------}\sin179^\circ=\frac{180}{2^{179}}$

$\frac{2^{180}}{45}\sin1^\circ\sin2^\circ-\text{------}\sin179^\circ=2^3=8$

C. we know that $\frac{z_3-z_2}{z_1-z_2}=\frac{|z_3-z_2|}{|z_1-z_2|}.e^{\frac{i\pi}{2}}$

$\Rightarrow\quad z_3-z_2=i(z_1-z_2)\qquad\Rightarrow(z_3-z_2)^2+(z_1+z_2)^2=0$

D. $z=\frac{-1}{2}i\left(1+i\sqrt{3}\right)=i\omega^2\qquad z^{101}=i\omega$

$\left(z^{101}+i^{100}\right)^{106}=-\left(i\omega^2\right)^{106}=-\omega^2$

$\therefore\quad -\omega^2=\left(i\omega^2\right)^{2n}=i^\pi\omega^{2n}\,\omega^{2\pi-2}i^\pi=-1$

This is possible only when N + 4r + 2 and 2x -2 is multiple of 3 i.e., 2 (4r + 2) – 2 is multiply of 3

i.e., 8r + 2 is multiple of $3\Rightarrow r=2\quad\therefore x=10\quad\therefore\frac{2}{5}k=4$

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PHYSICS

18. $A \propto \frac{1}{r}$

19. $\bar{L} = I_1\omega_1\hat{e}_1 + I_2\omega_2\hat{e}_2 + I_3\omega_3\hat{e}_3$

20. $\frac{q^2}{2C} = 3J$ Final energy = $\frac{\left(\frac{q}{2}\right)^2}{2C} \times 2 = 1.5J$

21. $\phi = \theta_m \sin\left\{2t - (3x - 4y) + \frac{\pi}{3}\right\}$

$\hat{n}$ will be along the normal to plane 3x – 4y = c $\hat{n} = \frac{3\hat{i} - 4\hat{j}}{5}$

In general ax + by + cz + d = 0 $\hat{n}$ will be along $a\hat{i} + b\hat{j} + c\hat{k}$

Wave = speed = $\frac{w}{k} = \frac{2}{5} = 0.4\,m/s$

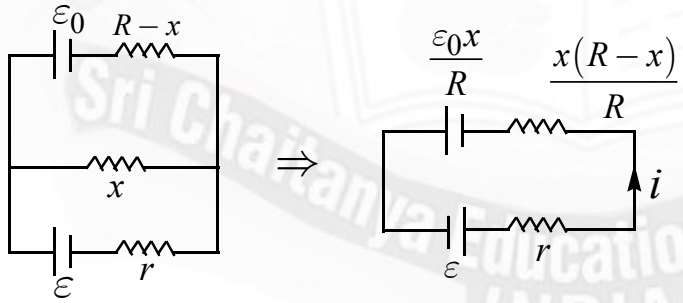
22. Both the particles are performing circular motion in C – frame with angular speed $\omega = \frac{u}{\ell}$, the CM moves uniformly with velocity $\frac{u}{2}$. Radius of circular path is $\frac{\ell}{2}$.

Tension remains constant equal to $m\left(\frac{\ell}{2}\right)\left(\frac{u}{\ell}\right)^2 = \frac{mu^2}{2\ell}$.

In G – frame trajectory of both the particles is cycloid, which is identical to trajectory of a particle on circumference of purely rolling ring of radius R which travels a distance 8R is in one time period.

23. $I' = I$ $\bar{B}' = \bar{B}$ $\bar{E}' = \bar{V} \times \bar{B}$

24. For Case–II



$i = \frac{\varepsilon + \frac{\varepsilon_0 x}{R}}{r + \frac{x(R-x)}{R}} = \frac{\varepsilon R + \varepsilon_0 x}{Rr + Rx - x^2}$: i increases with x for 0 < x < R, hence deflection of galvanometer increases as the jockey is moved from A to B.

25. $a = \frac{Mg\sin\beta + mg\sin\alpha.\cos(\alpha - \beta)}{M + m\sin^2\alpha} = 8\,m/s^2$

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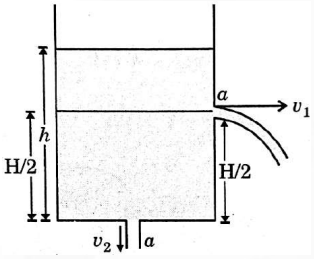
26. $F_1 \rightarrow \boxed{\quad\quad} \rightarrow F_2$ (ℓ, A, Y)

$$\Delta\ell = \frac{\ell}{AY}\left\{\frac{F_2 - F_1}{2}\right\} = \frac{4}{\left(10^{-4}\right)\left(2\times10^{8}\right)} \times \frac{100}{2} = 1\times10^{-2}\,m$$

27. $\Sigma\vec{F}_i \cdot d\vec{r}_i = 0 \Rightarrow -Tdy_A + Tdy_B - Pdx_F = 0 \Rightarrow T = \left(\frac{a+b}{a}\right)P\tan\theta$

28. $\frac{hC}{\lambda} = \phi + K \qquad \lambda_d = \frac{h}{\sqrt{2mK}}$

29. $v_1 = \sqrt{2g\left(h - \frac{H}{2}\right)} \quad v_2 = \sqrt{2gh}$



By continuing equation

$$A\left(-\frac{dh}{dt}\right) = a\left(v_1 + v_2\right) \quad A\left(-\frac{dh}{dt}\right) = a\left\{\sqrt{2g\left(h - \frac{H}{2}\right)} + \sqrt{2gh}\right\} \quad t = \frac{2A}{3a}\left(\sqrt{2}-1\right)\sqrt{\frac{H}{g}}$$

30. $F_{net} = m\omega^2 r$

31. f = 3 A; C= $\frac{3}{2}R + \frac{R}{3}$ B; C = $\frac{3}{2}R - \frac{R}{3}$ C; $C_v + 2R$

Process equation for Option 3. $\frac{\cancel{R}}{1-a} = 2\cancel{R}$ $1 - a = \frac{1}{2}$

$a = \frac{1}{2}$ $PV^{1/2} = \text{constant}$

D; $C = C_v - 2R$ Process equation for Option D.

$\frac{\cancel{R}}{1-a} = 2\cancel{R}$ $1 - a = \frac{1}{2}$ $a = \frac{3}{2}$ $PV^{3/2} = \text{constant}$

32. When current in increased or, loop is moved towards the wire, flux increases, loop is repelled and vice versa.

33. Alpha decay $\rightarrow$ Z decreases by 2.
Positron emission $\rightarrow$ Z decreases by 1 (lighter nuclei)
Proton emission $\rightarrow$ Z decreased by 1 (heavier nuclei)
Nuclear fission $\rightarrow$ Z is nearly halved.

34. For a mirror $\frac{dv}{dt} = -m^2\frac{du}{dt}$

For a lens $\frac{dv}{dt} = m^2\frac{du}{dt}$

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CHEMISTRY

35. Step 1: Determine empirical formula.
Moles of CO_2 = 0.330 / 44 = 0.00750 → C = 0.00750 mol.
Mass C = 0.090 g.
Moles of H_2O = 0.135 / 18 = 0.00750 → H = 0.01500 mol.
Mass H = 0.015 g.
$N\,mass = 0.150 \times 0.21 = 0.0315g \rightarrow 0.00225\,mol$.
O by difference = 0.0135 g → 0.00084375 mol.
Ratio $\approx C_5H_{11}NO_2 \rightarrow$ Valine.
Step 2: pI = (2.20 + 9.80) / 2 = 6.00.
Correct Option: (C) Valine, pI = 6.

36. Identification of orbitals:
(i) is the 1s orbital (no radial node, l = 0).
(iii) has factor $(2 - r/a_0)$ and exponential $e \wedge (-r/2a_0)$, characteristic of the 2s orbital (one radial node, l = 0).
(ii) is a 3s-type orbital (polynomial of degree 2 in σ with l = 0).
(iv) has a factor σ(4 − σ), typical of a 3p-type orbital (l = 1).
(A) For orbital (iv): $\psi_r \propto \sigma(4-\sigma)$.
The zeros of this polynomial are $\sigma = 0$ and $\sigma = 4$. $\sigma = 0$ corresponds to r = 0 (the origin), and σ = 4 corresponds to a single non-zero finite value of r. Therefore ψ_r vs r intersects the axis only once at a non-zero r. Statement (A) is FALSE.
(B) For orbital (i): $\psi_r = 2(1/a_0) \wedge (3/2) e \wedge (-r/a_0)$.
The radial probability density is $P(r) = 4\pi r^2 |\psi_r|^2 \propto r^2 e \wedge (-2r/a_0)$.
dP/dr = 0 gives $r = a_0$ as the position of maximum probability. Hence statement (B) is TRUE.
(C) Orbital (ii) is an s-orbital (l = 0). The number of angular nodes equals l.
Thus the number of angular nodes is 0, so there is no angular node. Statement (C) is TRUE.
(D) For orbital (iii): $\psi_r \propto (2 - r/a_0)$
Setting the factor equal to zero gives $2 - r/a_0 = 0 \Rightarrow r = 2a_0$.
This is the position of the radial node from the origin. Statement (D) is TRUE.

37. There are tetrahedral voids.

38. $BF_4^-, NH_4^+, CH_4, SiH_4^-$ all are tetrahedral

39. NCERT

40. Primary alchol and aldhyde and ketone are present

41. A) Adsorption decreases residual surface forces; does not increase them.
B) Adsorption depends on surface area; changing surface area changes adsorption.
C) Entropy decrease of system is NOT equal to entropy increase of surroundings.
D) Adsorption is enthalpy-driven (exothermic), not entropy-driven.

42. For the two sparingly soluble salts:
$Ag_2C_2O_4(s) \rightleftharpoons 2Ag^+ + C_2O_4^{2-}; \qquad K_{sp1} = 1.25 \times 10^{-11}$

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$Ag_2CO_3(s) \rightleftharpoons 2Ag^+ + CO_3^{2-},\ K_{sp2} = ?$

Initial moles of $K_2C_2O_4 = 1.485$

At equilibrium, 0.36 mol of K_2CO_3 is present, so

$n(K_2C_2O_4)_{eq} = 1.485 - 0.36 = 1.125\,mol$

Hence, using same degree of dissociation,

$$\frac{\left[C_2O_4^{2-}\right]}{\left[CO_3^{2-}\right]} = \frac{1.125}{0.36} = 3.125$$

$$\frac{K_{sp2}}{K_{sp1}} = \frac{\left[CO_3^{2-}\right]}{\left[C_2O_4^{2-}\right]} = \frac{1}{3.125} = 0.32$$

$$K_{sp2} = 1.25\times10^{-11}\times0.32 = 4\times10^{-12}\,mol^3L^{-3}$$

Thus, $K_{sp} = x\times10^{-y}$ with $x = 4, y = 12$, so $\frac{y}{x} = \frac{12}{4} = 3$

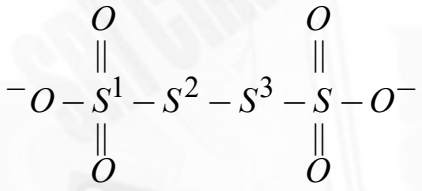
43. $15.5\ ^0C$

44. $\underset{X}{5I^-} + IO_3^- + 6H^+ \rightarrow \underset{Z}{3I_2} + 3H_2O$

From NaI

$Na_2SO_3 + S \rightarrow \underset{A}{Na_2S_2O_3}$

$Na_2S_2O_3 + I_2 \rightarrow \underset{X}{2NaI} + \underset{X^1}{Na_2S_4O_6}$



In the above structure 1 and 3 bonds are of equal length.

45. $(S)+(T), 0-5^\circ C, mild\ H^+;$

Diazonium salt (S) couoples with aromatic amine (T) to form an azo dye:

$O_2N - C_6H_4 - N = N - C_6H_4 - NH_2$

This is compound (U).

46. All compounds liberate oxygen.

47. $(1-a)\times6900 + a\times2900 = 3900 \Rightarrow a = \frac{3}{4}$

48. (a) Cyclohexanol $\rightarrow$ cyclohexanone

Secondary alcohol $\rightarrow$ ketone can be done by Cu/573 K(q) (dehydrogenation) or PCC (s) (mild oxidation)

$\Rightarrow a \rightarrow q, s$

(b) 1 – Methylcyclopenene 2-methylcyclopentanol

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The OH ends up on the less substituted carbon of the double bond $\rightarrow$ anti-Markovnikov hydration of an alkene, i.e., hydroboration-oxidation

$B_2H_6 / THF; H_2O_2 / OH^- (r)$.

$\Rightarrow c \rightarrow q, s$

c) $CH_3 - CH = CH - CH_2 - CH_2OH \rightarrow$ conjugated dienal

Primary allylic alcohol $\rightarrow$ conjugated aldehyde can be formed either by Cu/573K (q) (dehydrogenation/oxidation) or by PCC (s) (allylic oxidation).

$\Rightarrow c \rightarrow q, s$

d) Vicinal diol (pinacol) $\rightarrow$ ketone (pinacolone)

This is the classic pinacol – pinacoloe rearrangement, which uses conc. $H_2SO_4 (p)$.

$\Rightarrow d \rightarrow p$

49. A) Graphite has layered lattice structure in which carbon atoms are involved in sp^2 hybridization bond order is 1.33 can be used as solid lubricant and a good conductor of electricity.

B) Boron nitride has same structure and properties as that of graphite except it is not conductor of electricity.

C) Fullerenes contain sp^2 carbons and are electrical conductors.

D) Borinc acid contain sp^2 boron atoms layered lattice structure can be used as solid lubricant.

50. P–1,2,3, Q-1, R–2,3, S–1,5

51. Lead and copper are extracted by self reduction starting from their sulphide ores. Lead can also be extracted by carbon reduction. Silver is extracted by hydrometallurgy.

Extraction of boron involves decomposition of its $iodide \left(BI_3 \xrightarrow{\Delta} 2B + 3I_2 \right)$.

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